

Variables in Aggregation Expressions

Aggregation expressions can use both user-defined and system variables.

Variables can hold any BSON type data. To access the value of the variable, prefix the variable name with double dollar signs (\$\$); i.e. "\$\$<variable>".

If the variable references an object, to access a specific field in the object, use the dot notation; i.e. "\$\$<variable>.<field>".

User Variables

User variable names can contain the ascii characters `[_a-zA-Z0-9]` and any non-ascii character.

User variable names must begin with a lowercase ascii letter `[a-z]` or a non-ascii character.

System Variables

MongoDB offers the following system variables:

Variable	Description
<code>NOW</code>	A variable that returns the current datetime value. <code>NOW</code> returns the same value for all members of the deployment and remains the same throughout all stages of the aggregation pipeline.
<code>CLUSTER_TIME</code>	A variable that returns the current timestamp value. <code>CLUSTER_TIME</code> is only available on replica sets and sharded clusters. <code>CLUSTER_TIME</code> returns the same value for all members of the deployment and remains the same throughout all stages of the pipeline.
<code>ROOT</code>	References the root document, i.e. the top-level document, currently being processed in the aggregation pipeline stage.
<code>CURRENT</code>	References the start of the field path being processed in the aggregation pipeline stage. Unless documented otherwise, all stages start with <code>CURRENT</code> the same as <code>ROOT</code>. <code>CURRENT</code> is modifiable. However, since <code>\$<field></code> is equivalent to <code>\$\$CURRENT.<field></code>, rebinding <code>CURRENT</code> changes the meaning of <code>\$</code> accesses.
<code>REMOVE</code>	A variable which evaluates to the missing value. Allows for the conditional exclusion of fields. In a <code>\$project</code>, a field set to the variable <code>REMOVE</code> is excluded from the output. For an example of its usage, see Conditionally Exclude Fields.
<code>DESCEND</code>	One of the allowed results of a <code>\$redact</code> expression.
<code>PRUNE</code>	One of the allowed results of a <code>\$redact</code> expression.

Variable	Description
KEEP	One of the allowed results of a <code>\$redact</code> expression.
SEARCH_META	A variable that stores the metadata results of an Atlas Search query. In all supported aggregation pipeline stages, a field set to the variable <code>\$\$SEARCH_META</code> returns the metadata results for the query. For an example of its usage, see Atlas Search facet and count.
USER_ROLES	Returns the roles assigned to the current user. For use cases that include <code>USER_ROLES</code>, see the <code>find</code>, <code>aggregation</code>, <code>view</code>, <code>updateOne</code>, <code>updateMany</code>, and <code>findAndModify</code> examples. <i>New in version 7.0.</i>

TIP

See also:

- `$let`
- `$redact`
- `$map`